

TMM4128:

Machine Learning for Engineers

Andrei Lobov

e-mail: andrei.lobov@ntnu.no

phone: +47 45 913 455

room: 213, Verkstedteknisk

web: <http://www.lobov.biz/academia>

Slides:> <http://lobov.biz/academia/mle/250411>

Outline : Concluding remarks

- What did we do?
- Grading
- What to expect on the exam?
- Q&A session in mid-May (20.05.)
- What's next?



What did we do?

ToC : Lectures and Seminars

- L1: [MLE Course housekeeping / ML project checklist](#) (16.01.)
- L2: [Tools, setting up development environment](#) (17.01.)
- L3: [Classification](#) (23.01.)
- L4: [Training models](#) (24.01.)
- L5: [Support Vector Machine](#) (30.01.)
- L6: [Decision trees](#) (31.01.)
- L7: [Ensemble Learning and Random Forests](#) (06.02.)
- L8: [Recap of ML tools](#) (07.02.)
- L9: [Dimensionality reduction](#) (13.02.)
- L10: [Unsupervised learning techniques, Part I](#) (14.02.)
- L11: [Unsupervised learning techniques, Part II](#) (20.02.)
- L12: [Artificial Neural Networks](#) (21.02.)
- L13: [Artificial Neural Networks \(cont.\) and A2 description](#) (27.02.)
- L14: [Training Deep Neural Networks](#) (13.03.)
- L15: [Training DNN \(cont.\)](#) (14.03.)
- L16: [Custom models and training with TensorFlow](#) (20.03.)
- L17: [Convolutional Neural Networks](#) (21.03.)
- L18: [Convolutional Neural Networks \(cont.\)](#) (27.03.)
- L19: [Reinforcement learning {and KBE}](#) (28.03.)
- L20: [Reinforcement learning {TF support}](#) (03.04.)
- L21: [Reinforcement learning {TF support} \(DQN\)](#) (04.04.)
- L22: [Reinforcement learning {TF support} cont.](#) (10.04.)
- L23: Concluding remarks (11.04. - this presentation)

Grading

A: 89-100

B: 77-88

C: 65-76

D: 53-64

E: 41-52

F: 0-40

Exam gives **60% of the course grade**.

Assignments give **40% of the course grade**.

Both exam and **all** the assignments have to be passed (not “F”), to be able to pass (above “F”) the course. A1 (10 points max) + A2 (10 points max) + A3 (5 points max) can give maximum of 25 points. This is then translated to 0..100 scale and, finally, to the A-F grade. Exam grade (A-F) will have a bigger effect than assignment grade (A-F).

Exam data and location

Check for the INFO:

[Examination tab](#)

Aids: [Code D](#) /

No printed or hand-written support material is allowed. A specific basic calculator is allowed.

Place and room for school exam

The specified room can be changed and the final location will be ready no later than 3 days before the exam. You can find your room location on Studentweb.

Idrettssenteret (Dragvoll)

[Map to building](#)

Room Storhall del 2

50 candidates

TMM4128

Machine Learning for Engineers

anical
eering

Choose study year

Study year 2024/2025

Credits

7.5

Course start

Spring 2025

Language of instruction

English

Examination arrangement

Aggregate score

Level

Second degree level

Duration

1 semester

Location

Trondheim

About

Timetable

Examination

Examination

Examination arrangement: Aggregate score

Grade: Letter grades

Ordinary examination - Spring 2025

School exam

Weighting	60/100
Examination aids	Code D
Date	2025-05-28
Time	09:00
Duration	4 hours
Exam system	Paper

What to expect on the exam?

- 28.05., 09:00
- Location (Check 3 days before the exam for the final information)
- Pencil and paper exam.
- Max 1 A4 page to answer a question.
- 5 questions.

Supervised / Unsupervised / Reinforcement learning methods and tools. ML project steps.

Sample questions

1. What is a difference between precision and recall? When one can be preferable than another? How to decide on a good combination of those?
2. Why does an ensemble learning work? How the weak learners can be turned into strong learners?
3. Receptive field in CNN: What is its purpose? What is a difference between SAME and VALID padding? Can an application of these paddings result, correspondingly, in a different number of neurons in a given layer with respect to a previous layer? - Why, if yes, or why, if not?

When answering exam questions

Explain interconnections between 'elements' in your answer.

You can ask yourself “How to...” questions. An architecture? - how the blocks get connected in it. A language/representation? - how this is being used. An algorithm? - how it is being applied... Hyperparameters - what is their meaning? How changing them may affect the learning?

Q&A session in mid-May : Tue. 20.05.25, 14:00-15:00

Mark your calendar! Tuesday, 20.05.2025, 14:00-15:00

Microsoft Teams:

>>> [Join the meeting online, Tue. 20.05.2025, 14:00-15:00](#)

Questions?

Andrei Lobov

andrei.lobov@ntnu.no

+47 45 913 455

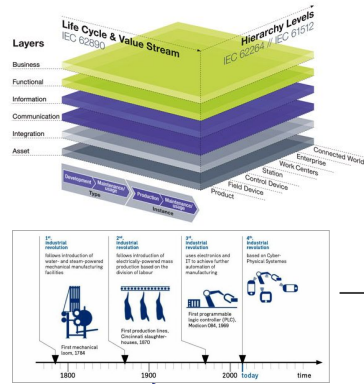
What is next?

Some options:

- TMM4270 (Autumn 2025)
- On the role of the human
- Towards “digital engineering brain”

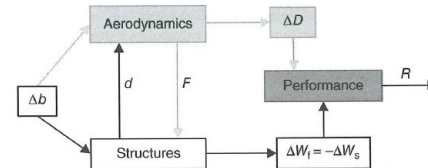
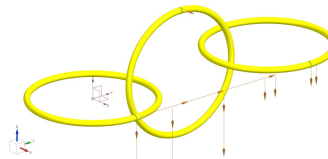
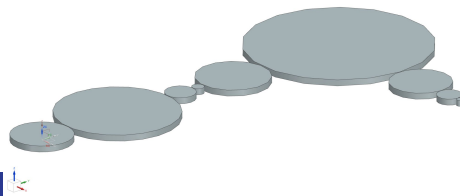
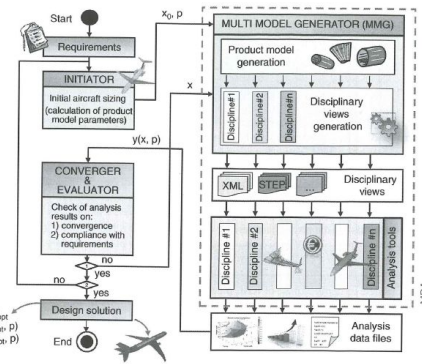
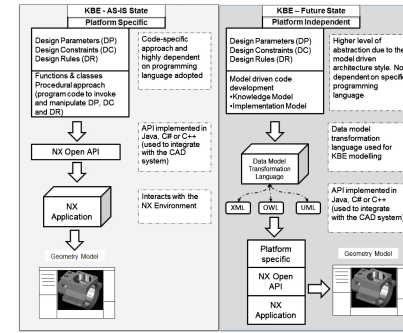
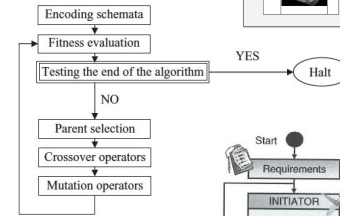
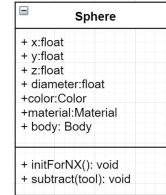
TMM4270 : Knowledge Based Engineering, Introduction (Autumn 2025)

Languages and tools to support KBE development.



```
(child) max_ext_disk:
{
    class, ug_cylinder;
    diameter, max_ext_dia;;
    height, max_ext_length;;
    Origin, Point(0,0,0);
    Direction, Vector(0,0,1);
}
```

```
SELECT
(xsd:float(?h)*xsd:float(?r)*xsd:float(?r)*a
fn:pi() AS ?volume) ?cyl
WHERE
{
    ?cyl a kbe:Cylinder.
    ?cyl kbe:hasRadius ?r.
    ?cyl kbe:hasHeight ?h.
}
```



On the role of human

Some speculative thoughts

Limits of Logic

<https://www.youtube.com/watch?v=V9ohtKameio>

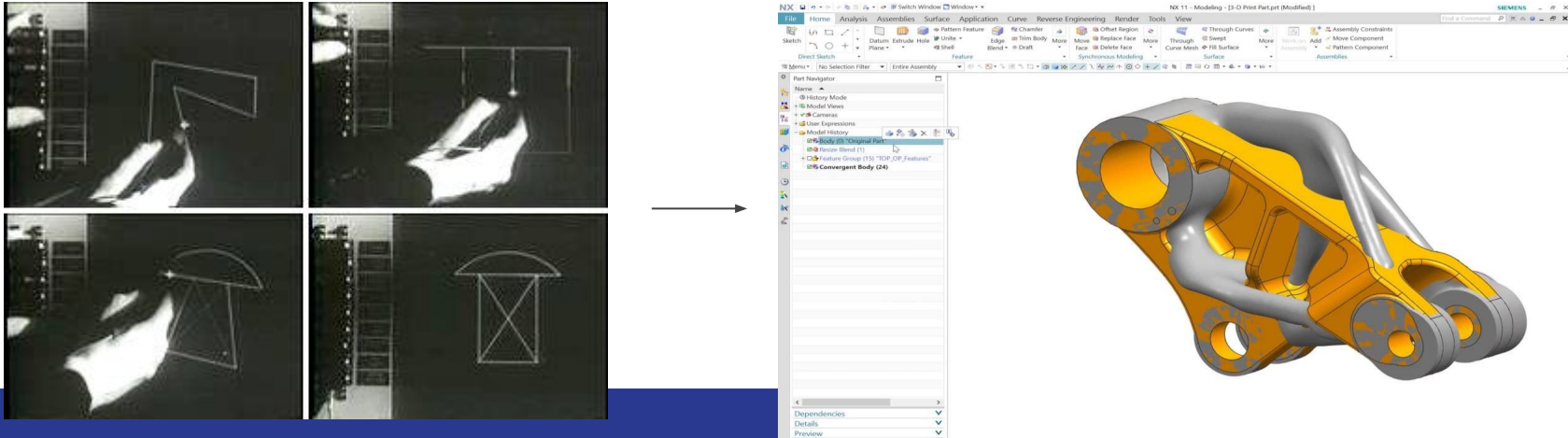
- (1) $\forall a: \forall b: (a + Sb) = S(a + b)$
- (2) $\forall b: (S0 + Sb) = S(S0 + b)$
- (3) $(S0 + S0) = S(S0 + 0)$
- (4) $\forall a: (a + 0) = a$
- (5) $(S0 + 0) = S0$
- (6) $S(S0 + 0) = SS0$
- (7) $(S0 + S0) = SS0$



Towards “digital engineering brain”

Computer-aided Design (CAD) - Background

- CAD used to model product geometries.
- May record also some non-geometric design specs.
- Design model may serve as the interface between design and manufacturing.
- CAD is critical for Computer-aided manufacturing (CAM).
- Development of CAD began in 1960s ...



Network / Value chain / B2B / B2C / C2B ...

ERP

PLM

MES

Process model

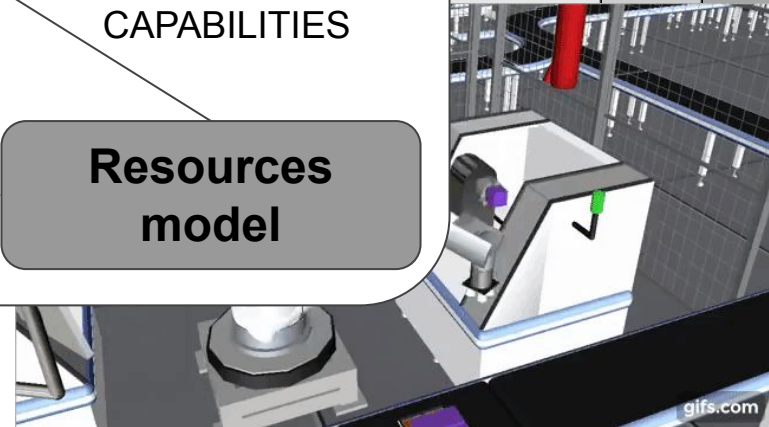
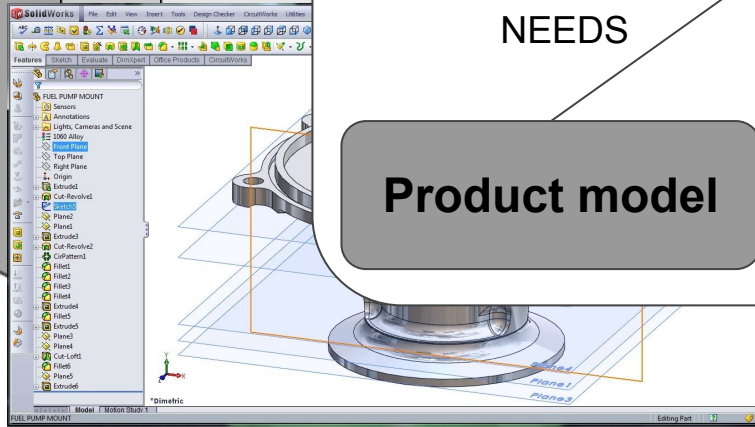
NEEDS

CAPABILITIES

Product model

MATCH?

Resources model



If not, how to make it possible?

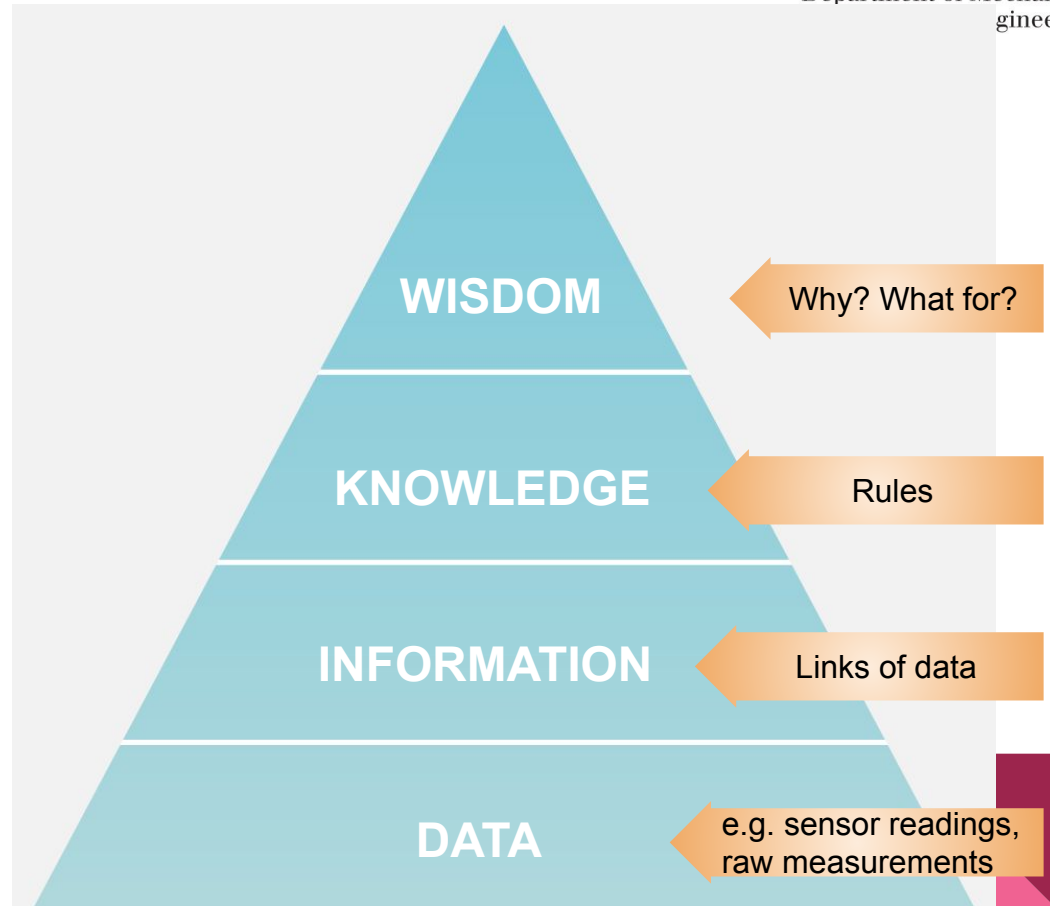
What is a knowledge?

The DIKW pyramid perspective

DIKW definition in 1989



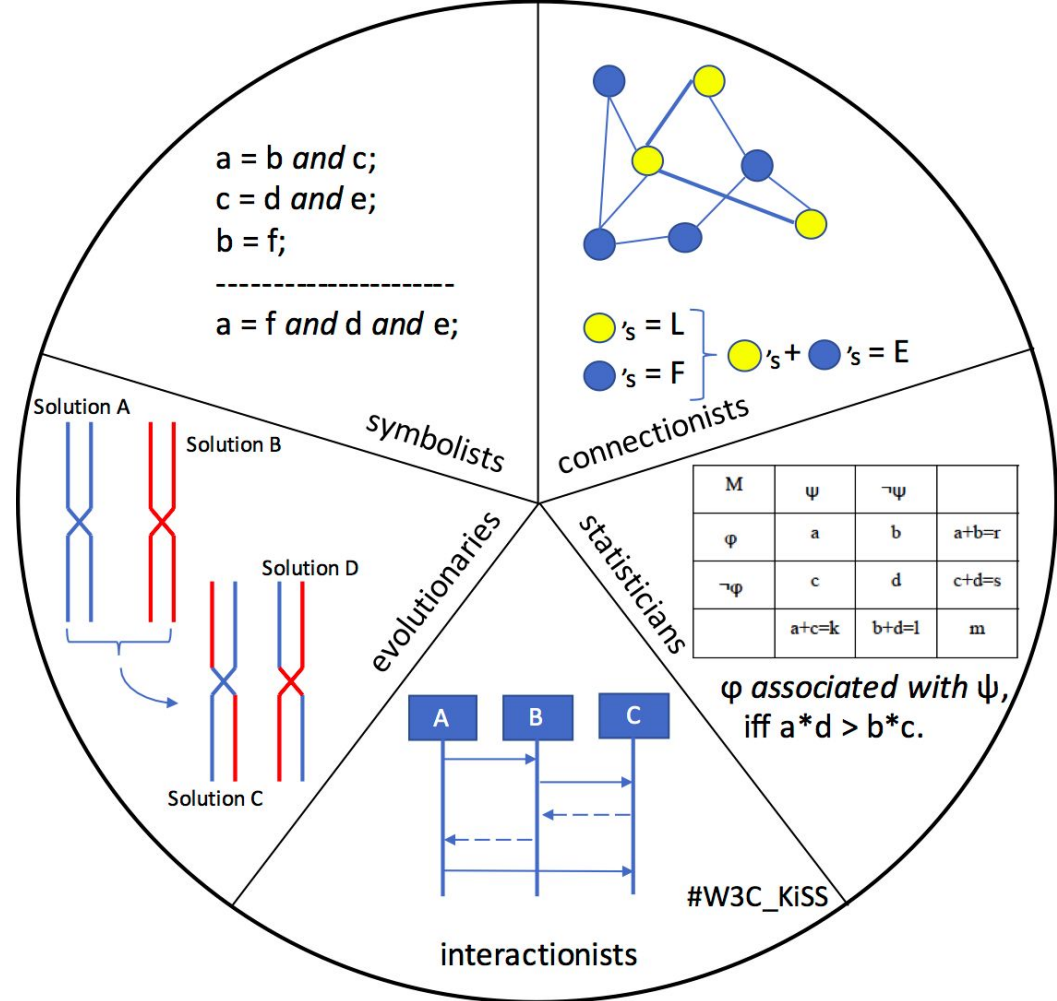
Russell Ackoff (1919 – 2009)



**CAN WE CLIMB TO THE TOP?
(TO BUILD INTELLIGENT SYSTEMS)**

5 schools of thought : Encoding the rules

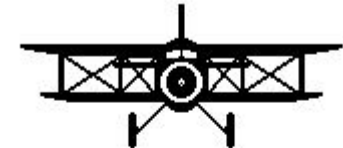
- Symbolists
- Connectionists
- Evolutionaries
- Statisticians
- Interactionists



Boundaries for evolution / design automation to keep in mind?

“An aircraft optimization starting with a biplane could evolve into a monoplane with a low, mid, or high wing; but a biplane will not arise from a configuration initialized as monoplane.”

[source: Sobieszczanski-Sobieski, Morris and van Tooren, “Multidisciplinary Design Optimization Supported by Knowledge Based Engineering”, Wiley, ISBN 9781118492123, 2015]



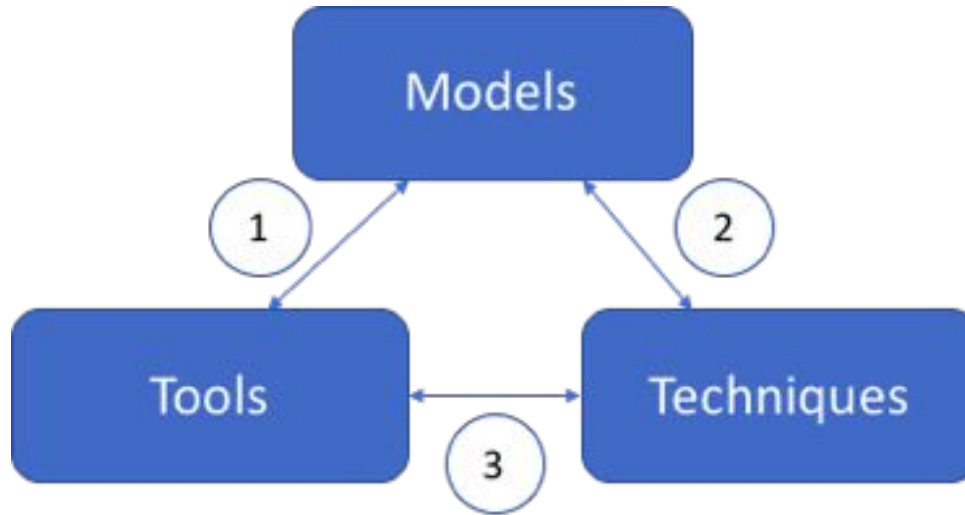
BIPLANE



MONOPLANE

Models-Tools-Techniques (1)

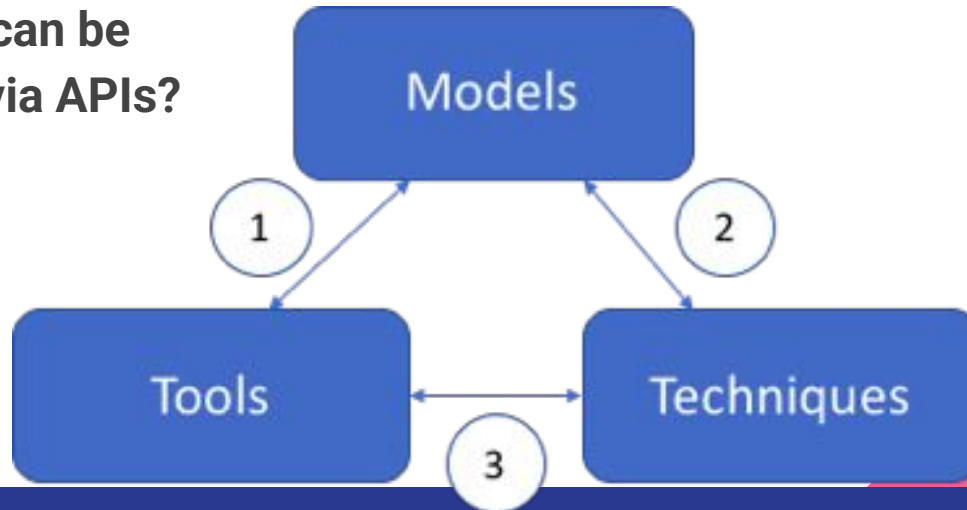
Systematic development may require a methodology that can be seen as a triangle:



J. W. Satzinger, R. B. Jackson and S. D. Burd, Object-Oriented Analysis and Design with Unified Process, Thomson, 2005, ISBN 0-619-21643-3

Models-Tools-Techniques (2)

- **Models** to be able to grasp adequately the problem(s).
- **Tools** to provide flawless use.
- **Techniques** to tell when, what, how and why to use.
- **How mature these with respect to each other?**
- **How these can be integrated via APIs?**



ChatGPT “routing” a pipe?

Please route a pipe from coordinate (10, 10, 12) in 3D space to the point (12, 45, 300). What is a length of such a pipe?

To find the length of the pipe that routes from coordinate (10, 10, 12) to the point (12, 45, 300) in 3D space, we can use the distance formula between two points in 3D space. The distance formula is:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Where:

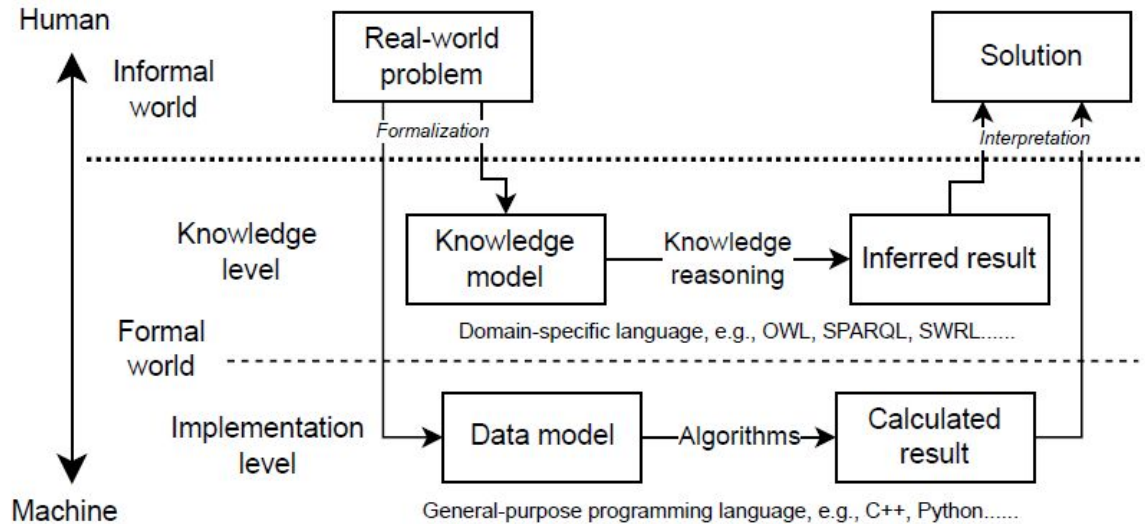
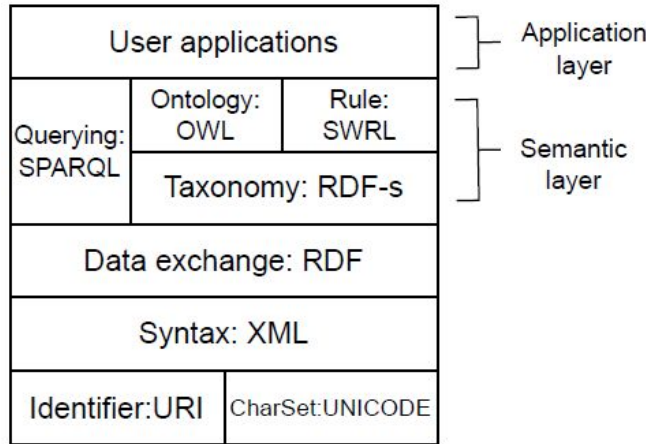
- (x_1, y_1, z_1) = Starting coordinate (10, 10, 12)
- (x_2, y_2, z_2) = Ending point (12, 45, 300)

Plugging in the values:

$$\begin{aligned} \text{Distance} &= \sqrt{(12 - 10)^2 + (45 - 10)^2 + (300 - 12)^2} \\ \text{Distance} &= \sqrt{2^2 + 35^2 + 288^2} \\ \text{Distance} &= \sqrt{4 + 1225 + 82944} \\ \text{Distance} &= \sqrt{84173} \\ \text{Distance} &\approx 290.08 \end{aligned}$$

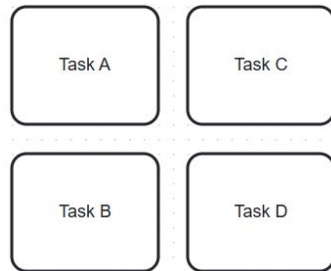
So, the length of the pipe from coordinate (10, 10, 12) to the point (12, 45, 300) is approximately 290.08 units.

Digital transformation through semantic modelling (1)



Digital transformation through semantic modelling and Business Process Model and Notation (BPMN)

- To document business processes.
- To improve communication between business analysts, developers, and stakeholders.
- To support process automation by dynamically composing and executing the models.



Generating BPMN with LLM:

Generate BPMN 2.0 file for CAD creating a 1 meter long pipe of external diameter 30 cm and the wall thickness 5 cm.

X Cad Pipe Bpmn

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <definitions xmlns="http://www.omg.org/spec/BPMN/20100524/MODEL"
3   xmlns:bpmndi="http://www.omg.org/spec/BPMN/20100524/DI"
4   xmlns:omgdc="http://www.omg.org/spec/DD/20100524/DC"
5   xmlns:omgdi="http://www.omg.org/spec/DD/20100524/DI"
6   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
7   xsi:schemaLocation="http://www.omg.org/spec/BPMN/20100524/MODEL BPMN20.xsd"
8   id="Definitions_1"
9   targetNamespace="http://bpmn.io/schema/bpmn">
10
11 <process id="Process_Create_Pipe" name="Create 1m Pipe in CAD" isExecutable="true">
12   <startEvent id="StartEvent_1" name="Start"/>
13
14   <task id="Task_Define_Parameters" name="Define Pipe Parameters"/>
15
16   <task id="Task_Create_3D_Model" name="Create 3D Pipe Model">
17     <incoming>SequenceFlow_1</incoming>
18     <outgoing>SequenceFlow_2</outgoing>
19   </task>
```

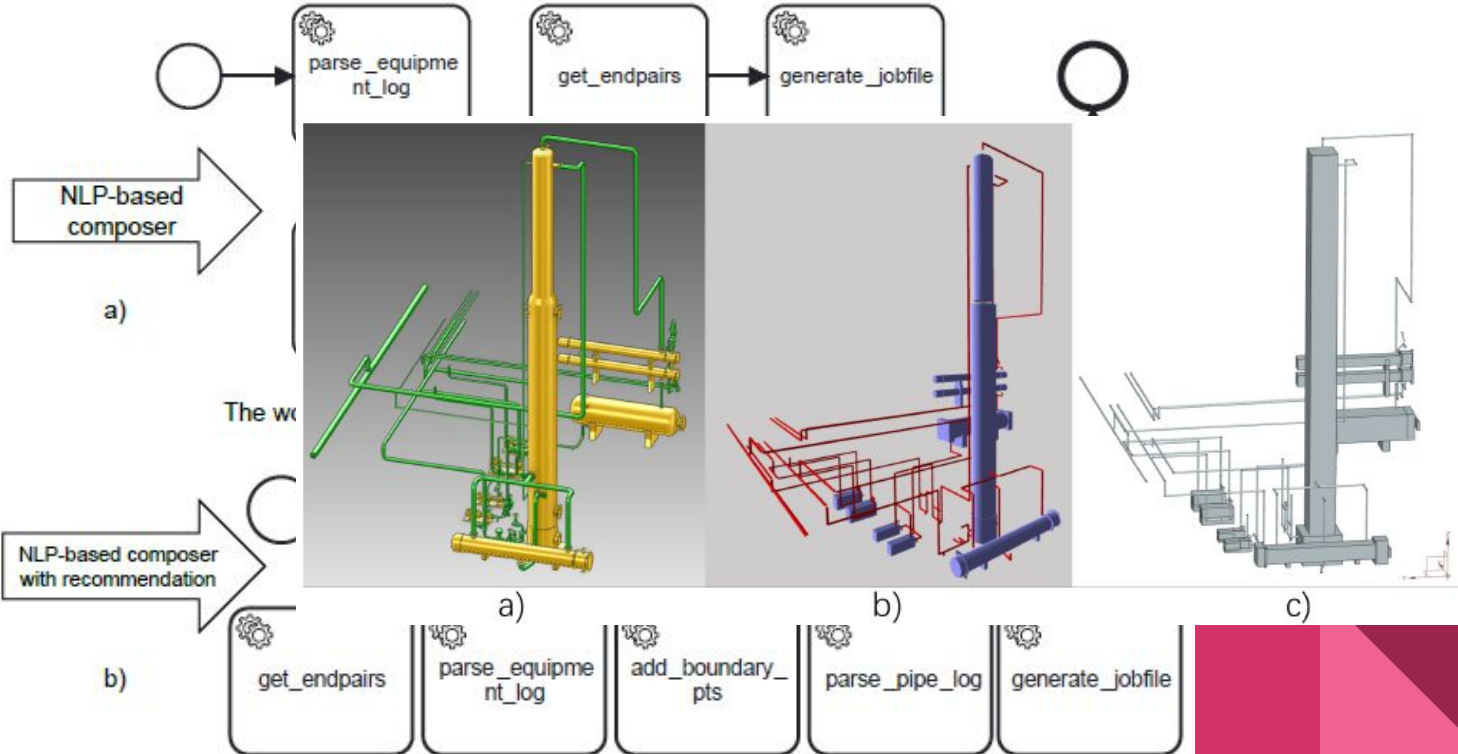
To try: <https://demo.bpmn.io/s/start>

Digital transformation through semantic modelling (3)

The workflow can be executed after configuration.

Workflow description in NL text (high quality):

"KB, design": "Parse the equipment log file, then add boundary points to these equipment elements, parse the pipe log file, and generate the end pair file, then generate a job file, use pipe router to route pipes, then upload the pipe elements to KB."



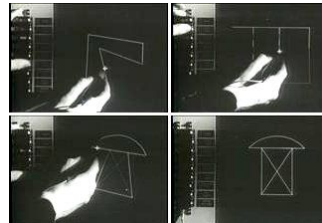
Workflow description in NL text (low quality):

"KB, design": "route pipes, then upload the routed pipe to KB."

Takeaways



- There are standards and languages (representation and programming) available to allow data, information, knowledge representation and processing for various application domains.
- Learning or retrieving knowledge without being explicitly programmed/hardwired has already some successful ad hoc solutions.
- Further efforts are required to push the boundaries for automatically forming and executing “engineering recipes” - the tasks of a “digital engineering brain” (e.g. in the form of BPMN).



Generate BPMN 2.0 file for CAD creating a 1 meter long pipe of external diameter 30 cm and the wall thickness 5 cm.

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <definitions xmlns="http://www.omg.org/spec/BPMN/20100524/MODEL"
3   xmlns:bpmndi="http://www.omg.org/spec/BPMN/20100524/DI"
4   xmlns:camdc="http://www.omg.org/spec/CD/20100524/DC"
5   xmlns:camdi="http://www.omg.org/spec/CD/20100524/DI"
6   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
7   xsi:schemaLocation="http://www.omg.org/spec/BPMN/20100524/MODEL BPMN20.xsd"
8   id="Definitions_1"
9   targetNamespace="http://bpmn.io/schema/bpmn">
10
11   <process id="Process_Create_Pipe" name="Create 1m Pipe in CAD" isExecutable="true">
12     <startEvent id="StartEvent_1" name="Start"/>
13
14     <task id="Task_Define_Parameters" name="Define Pipe Parameters"/>
15
16     <task id="Task_Create_3D_Model" name="Create 3D Pipe Model">
17       <incomingSequenceFlow id="incoming">
18         <outgoingSequenceFlow id="outgoing">
19         </tasks>
```